💰 Financial Operations Analytics Project Report

Revenue Forecasting, Churn & Profitability Analysis — U.S. Superstore

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SECTION 1: BUSINESS REQUIREMENTS DOCUMENT (BRD)

1.1 Executive Summary

Retail businesses operating across multiple product categories, customer segments, and geographic regions face a persistent challenge: understanding not just how much revenue they generate, but where they're actually making — or losing — money. A store can generate $2.3 million in sales and still operate with razor-thin margins if discounting is aggressive, if certain product lines are sold at a loss, or if specific regions consistently underperform.

This project analyzes 4 years of transactional data (2014–2017) from a U.S.-based retail superstore operating across all 50 states, selling Furniture, Office Supplies, and Technology to Consumer, Corporate, and Home Office segments. The dataset includes a critical field that most retail datasets lack: **actual profit per transaction** — enabling true profitability analysis, not just revenue tracking.

The project builds a comprehensive **Financial Operations Analytics Suite** with three integrated pillars: revenue analysis and forecasting, customer churn prediction, and profitability deep-dive (by product category, sub-category, region, state, and customer segment). The goal is to move leadership from reactive reporting ("what happened last quarter") to proactive intelligence ("where should we invest, cut, and intervene next quarter").

1.2 Business Problem Statement

Current State: The Superstore has grown to $2.3M in total sales across 4 years, serving 793 unique customers across 4 U.S. regions. However, gross profit is only ~$286K — a 12.5% profit margin — raising serious questions about the health of the business:

- **Discount Strategy Is Unclear:** The average discount is 15.6%, but the relationship between discounting and profitability is not well understood. Are discounts driving incremental volume that justifies the margin sacrifice, or are they simply eroding profit on sales that would have happened anyway?
- **Product Portfolio Has Hidden Losses:** Aggregate numbers mask the fact that entire sub-categories (Tables, Bookcases, Supplies) are operating at a net loss. Without sub-category-level P&L visibility, management cannot make informed decisions about product mix, pricing, or vendor negotiations.

- **Regional Performance Is Uneven:** Some states generate strong revenue but negative profit (e.g., Texas for Furniture, Ohio for Technology), while others are profitable but under-penetrated. There is no systematic framework for identifying where to invest vs. where to cut losses.
- **Customer Retention Is Unexplored:** With 793 customers over 4 years, the business depends on repeat purchasing. However, there is no quantified understanding of customer churn rate, churn drivers, or the lifetime value profile of different customer segments.
- **No Forward-Looking Capability:** Revenue and profit are tracked historically but there is no forecasting model to support quarterly or annual planning, hiring decisions, or inventory investment.

Desired Future State: A unified Financial Operations Analytics Suite that provides:

1. Revenue and profit trend analysis with decomposition and forward forecast
2. True profitability visibility at the sub-category, region, state, and segment level
3. Discount impact analysis — quantifying the relationship between discounting and profit
4. Customer churn prediction model with identified risk factors
5. An interactive executive dashboard that enables leadership to monitor, drill down, and act

1.3 Business Objectives

#	Objective	Success Metric
1	Decompose revenue and profit trends into underlying drivers (category, segment, region, discount)	Clear attribution of profit/loss to specific product lines, regions, and discount bands
2	Build a revenue forecast model for quarterly planning	3–6 month revenue forecast with quantified trend
3	Identify all loss-making sub-categories and quantify the magnitude of loss	Complete sub-category P&L with profit margin percentages
4	Analyze the relationship between discount levels and profitability	Discount-profit curve showing optimal discount thresholds
5	Quantify customer churn rate and identify top churn predictors	Churn rate calculated, top 5 predictors ranked by importance
6	Build a churn prediction model to flag at-risk customers	Classification model with AUC-ROC ≥ 0.70
7	Rank regions and states by profit efficiency (profit margin, not just revenue)	State-level profitability heatmap with margin analysis
8	Deliver an executive-grade interactive dashboard	5-page Tableau/Power BI dashboard covering all three analytical pillars

1.4 Stakeholders

Stakeholder	Role	Interest
CEO / General Manager	Executive sponsor	Overall business health, growth trajectory, strategic planning
CFO / Head of Finance	Revenue & profit owner	Margin analysis, discount strategy, forecasting for budgets
VP of Sales	Revenue growth	Regional performance, customer segment strategy, quota planning
VP of Marketing	Customer strategy	Churn reduction, customer segment targeting, campaign ROI
Merchandising / Category Management	Product strategy	Category and sub-category profitability, product mix optimization
Supply Chain / Logistics	Shipping & fulfillment	Ship mode cost-efficiency, regional logistics optimization
Data & Analytics Team	Technical delivery	Model development, dashboard maintenance, data pipeline

1.5 Scope

In Scope:

- Analysis of all transactions from the U.S. Superstore dataset (2014–2017)
- Revenue and profit trend analysis with monthly/quarterly decomposition
- Revenue forecasting using trend-based modeling
- Profitability analysis by category, sub-category, region, state, customer segment, and ship mode
- Discount impact analysis — quantifying the discount-to-profit relationship
- Customer churn definition, analysis, and binary classification modeling
- Customer segment comparison (Consumer, Corporate, Home Office)
- Shipping mode analysis and its relationship to profitability
- Interactive 5-page executive dashboard (Tableau / Power BI)
- Business recommendations with quantified financial impact

Out of Scope:

- Integration with live POS, ERP, or CRM systems
- Advanced time-series models (ARIMA, Prophet, LSTM) — noted as future enhancements
- Customer Lifetime Value (CLV) predictive modeling
- Vendor/supplier analysis or procurement optimization
- Inventory management or demand planning
- Marketing channel attribution or campaign-level analysis
- Real-time model deployment or API development
- Pricing optimization or dynamic pricing models

1.6 Assumptions & Constraints

Assumptions:

- Each Customer ID reliably identifies a unique customer across all transactions and years
- The "Profit" field represents true gross profit (revenue minus cost of goods sold) and is accurately recorded
- The "Sales" field represents the amount paid by the customer after discount (not list price)
- The "Discount" field represents the percentage discount applied (0.0 = no discount, 0.2 = 20% off)
- All transactions are denominated in U.S. dollars ($)
- The dataset represents the complete transaction history for this store (no missing orders)
- Ship Mode categories (Standard Class, Second Class, First Class, Same Day) have different cost structures, though shipping costs are not explicitly provided

Constraints:

- **No explicit shipping cost data:** Unlike the Olist dataset, the Superstore dataset does not include a separate freight/shipping cost field. Shipping cost impact must be inferred from Ship Mode and Profit columns.
- **No customer demographic data:** No age, gender, income, or household information. Customer analysis is purely behavioral.
- **Moderate dataset size:** 9,994 rows is sufficient for analysis and modeling but limits the granularity of sub-group analysis (e.g., state × sub-category × segment combinations may have very small sample sizes).

- **Fictional but realistic data:** This is a widely-used sample dataset (originally from Tableau's sample data). While not from a real company, it mirrors realistic retail dynamics and is recognized by recruiters and interviewers in the analytics space.
- **No marketing or advertising spend data:** Cannot directly connect customer acquisition cost to retention.
- **Time period (2014–2017):** Analysis methodology is timeless, but specific product trends may differ from current market conditions.

1.7 Key Business Questions This Project Will Answer

Revenue & Forecasting

1. **What is the Superstore's revenue and profit trajectory over 4 years?** Is growth accelerating or decelerating?
2. **What is driving revenue — order volume, average order value, or customer mix?**
3. **What is the projected revenue for the next 3–6 months?**
4. **Which months and quarters show the strongest seasonal patterns?**

Profitability

5. **Which sub-categories are profitable and which are operating at a loss?** What is the magnitude?
6. **What is the relationship between discount percentage and profit?** Is there a threshold where discounting becomes destructive?
7. **Which states generate the most revenue but the least profit?** Where are the margin leaks?
8. **How does profitability differ across customer segments (Consumer, Corporate, Home Office)?**
9. **Does Ship Mode choice correlate with profitability?**

Customer Churn

10. **What percentage of customers have churned (not purchased in the last 6–12 months)?**
11. **What are the top predictors of customer churn?**
12. **Can we build a model that flags at-risk customers before they churn?**
13. **How does customer lifetime behavior differ across segments?**

SECTION 2: PROJECT REQUIREMENTS DOCUMENT (PRD)

2.1 Data Source

Attribute	Detail
Dataset Name	Superstore Sales Dataset
Source	Kaggle
URL	https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
Alternative	Also bundled as sample data with Tableau Desktop
Records	9,994 transaction line items
Columns	21
Unique Orders	~5,009
Unique Customers	793
Geography	United States — all 50 states, 531 cities, 4 regions
Time Period	January 2014 – December 2017 (4 years)
Currency	U.S. Dollars ($)
Categories	3 (Furniture, Office Supplies, Technology)
Sub-Categories	17
Customer Segments	3 (Consumer, Corporate, Home Office)
Key Advantage	**Includes actual Profit per transaction** — enables true P&L analysis

2.2 Data Dictionary

Identification & Order Fields

Column	Data Type	Description	Notes
Row ID	Integer	Unique row identifier	Not analytically useful — can be dropped
Order ID	String	Unique order identifier (e.g., "CA-2017-152156")	One order can have multiple rows (line items). Format: Region-Year-Number

Column	Data Type	Description	Notes
Order Date	Date	Date the order was placed	Primary time dimension. Parse to datetime.
Ship Date	Date	Date the order was shipped	Used to calculate shipping speed: Ship Date – Order Date
Ship Mode	Categorical	Shipping method selected	4 values: Standard Class (~60%), Second Class (~19%), First Class (~15%), Same Day (~5%)

Customer Fields

Column	Data Type	Description	Notes
Customer ID	String	Unique customer identifier (e.g., "CG-12520")	**Consistent across orders** — suitable for customer-level analysis
Customer Name	String	Customer's full name	Not used in modeling. Could verify Customer ID uniqueness.
Segment	Categorical	Customer type/segment	3 values: Consumer (~52%), Corporate (~30%), Home Office (~18%)

Geographic Fields

Column	Data Type	Description	Notes
Country	String	Always "United States"	Single value — not analytically useful
City	String	Customer's city	531 unique cities
State	String	Customer's state	49 states represented (all except one)
Postal Code	Integer	Customer's zip code	631 unique codes
Region	Categorical	U.S. region	4 values: West (~32%), East (~29%), Central (~22%), South (~17%)

Product Fields

Column	Data Type	Description	Notes
Product ID	String	Unique product identifier	1,862 unique products
Category	Categorical	Product category	3 values: Office Supplies (~60% of rows), Furniture (~21%), Technology (~19%)
Sub-Category	Categorical	Product sub-category	17 values (see detailed breakdown below)
Product Name	String	Full product name	1,862 unique names. Can be used for product-level deep dives.

Sub-Category Breakdown:

Category	Sub-Categories
Furniture	Bookcases, Chairs, Furnishings, Tables
Office Supplies	Appliances, Art, Binders, Envelopes, Fasteners, Labels, Paper, Storage, Supplies
Technology	Accessories, Copiers, Machines, Phones

Financial Fields (Critical for This Project)

Column	Data Type	Description	Notes
Sales	Float	Revenue from the transaction ($)	This is the amount after discount. Range: $0.44 – $22,638
Quantity	Integer	Number of units sold	Range: 1–14
Discount	Float	Discount percentage applied	Range: 0.0 (no discount) to 0.8 (80% off). **Key profitability driver.**
Profit	Float	Gross profit from the transaction ($)	Range: –$6,599 to $8,399. **Can be negative** — this is the critical analytical feature.

Why Profit is transformative: Most publicly available retail datasets only include Sales (revenue). The inclusion of actual Profit enables true financial operations analysis: margin calculation, loss identification, discount optimization, and profitability-based segmentation. This is what makes the project title "Financial Operations" genuinely appropriate.

2.3 Data Quality Assessment Plan

Issue	Severity	Record Count	Resolution Strategy
No missing values	N/A	0 nulls across all 21 columns	No imputation needed — unusually clean dataset. Note this positively but don't skip profiling.
Duplicate rows	Low	~17 exact duplicates	Drop exact duplicates. Verify row count drops from 9,994 to ~9,977.
Country column is constant	Low	All rows = "United States"	Drop column — no analytical value.
Row ID is a sequential identifier	Low	All rows	Drop column — not predictive.
Customer Name contains PII	Low	All rows	Not used in modeling. Can be retained for lookup dashboards.
Negative Profit values	**Not an error**	~1,800 rows (~18%)	These are real losses — the most analytically valuable records in the dataset. Do NOT remove them.
High-discount transactions	Medium	Discounts up to 80%	Investigate whether extreme discounts (>50%) are clearance/liquidation events or data issues.
Order Date parsing	Low	All rows	Ensure consistent datetime format. Verify chronological range: Jan 2014 – Dec 2017.
Multi-line orders	**Critical**	~5,009 unique orders across 9,994 rows	One Order ID can have multiple rows (different products in same order). Must aggregate to order-level for customer and order-level analysis.

2.4 Feature Engineering Plan

Order-Level Derived Features

Feature	Derivation	Purpose
order_total_sales	SUM(Sales) grouped by Order ID	True order-level revenue
order_total_profit	SUM(Profit) grouped by Order ID	True order-level profitability
order_items_count	COUNT of rows per Order ID	Order size (number of line items)
order_avg_discount	AVG(Discount) per Order ID	Average discount applied to order
shipping_days	Ship Date – Order Date (days)	Fulfillment speed
profit_margin	Profit / Sales × 100	Margin percentage per transaction
is_profitable	1 if Profit > 0, else 0	Binary profitability flag
order_month	Year-month from Order Date	Monthly aggregation key
order_quarter	Year-quarter from Order Date	Quarterly aggregation key
order_year	Year from Order Date	Annual aggregation key

Customer-Level Derived Features (for Churn Modeling)

Feature	Derivation	Purpose
first_order_date	MIN(Order Date) per Customer ID	Customer acquisition date
last_order_date	MAX(Order Date) per Customer ID	Recency reference
order_count	COUNT(DISTINCT Order ID) per Customer ID	Purchase frequency
total_spent	SUM(Sales) per customer	Total revenue generated
total_profit	SUM(Profit) per customer	Total profit contribution
avg_order_value	total_spent / order_count	Spending intensity
avg_profit_margin	total_profit / total_spent × 100	Customer-level profitability
avg_discount	AVG(Discount) per customer	Price sensitivity

Feature	Derivation	Purpose
unique_categories	COUNT(DISTINCT Category) per customer	Engagement breadth
unique_subcategories	COUNT(DISTINCT Sub-Category) per customer	Product diversity
preferred_segment	MODE(Segment) per customer	Customer type (should be constant)
preferred_ship_mode	MODE(Ship Mode) per customer	Shipping preference
tenure_days	last_order_date – first_order_date (days)	Customer lifespan
recency_days	Reference date – last_order_date (days)	Time since last purchase
avg_shipping_days	AVG(shipping_days) per customer	Service experience
is_churned (target)	1 if recency_days > 365, else 0	**Binary churn label**

Churn threshold rationale (365 days): Unlike the Olist marketplace (where 90 days was appropriate for a platform with no inherent loyalty), this is a B2C/B2B retailer with 793 customers over 4 years. Many customers purchase quarterly or annually. A 365-day threshold ensures we're identifying truly inactive customers, not just seasonal buyers. This design choice should be discussed in interviews.

Sub-Category-Level Derived Features (for Profitability)

Feature	Derivation	Purpose
total_sales	SUM(Sales) per Sub-Category	Revenue
total_profit	SUM(Profit) per Sub-Category	True profitability
profit_margin_pct	total_profit / total_sales × 100	Margin efficiency
total_quantity	SUM(Quantity) per Sub-Category	Volume
avg_discount	AVG(Discount) per Sub-Category	Discount pressure
order_count	COUNT(DISTINCT Order ID) per Sub-Category	Transaction frequency
avg_order_value	total_sales / order_count	Unit economics
loss_transactions_pct	% of transactions where Profit < 0	Loss frequency

2.5 Analytical Approach

Pillar 1: Revenue Analysis & Forecasting

Phase	Analysis	Method	Deliverable
1A	Revenue and profit trends	Monthly/quarterly line charts (dual axis: sales + profit)	4-year trajectory with growth rate annotation
1B	Growth driver decomposition	Separate trends: order volume, AOV, customer count	Attribution of growth to specific levers
1C	Seasonal pattern analysis	Month-over-month and quarter-over-quarter patterns; holiday spikes	Campaign timing intelligence (November/December peaks)
1D	Category revenue trends	Stacked area chart: Furniture vs. Office Supplies vs. Technology	Category growth trajectory comparison
1E	Revenue forecasting	Linear regression trend + 3-month moving average	3–6 month forward projection with R^2 confidence

Pillar 2: Profitability Analysis (The Core Differentiator)

This is the pillar that makes this project genuinely a "Financial Operations" project — true P&L analysis that most portfolio projects cannot do.

Phase	Analysis	Method	Deliverable
2A	Sub-category P&L	Bar chart: Sales vs. Profit by sub-category (sorted by profit)	Identification of all 3 loss-making sub-categories (Tables, Bookcases, Supplies)
2B	Profit margin ranking	Horizontal bar: profit margin % by sub-category	Margin efficiency ranking — Copiers and Paper likely top, Tables worst
2C	Discount-Profit analysis	Scatter plot: Discount % vs. Profit; grouped analysis by discount band	**Critical insight:** At what discount threshold does profitability collapse?
2D	Regional profitability	Heatmap: State × profit margin; map visualization	Identify states that generate revenue but destroy profit
2E	Segment profitability	Comparison of Consumer vs. Corporate vs. Home Office on sales, profit, margin	Determine which segment is most profitable per dollar of sales
2F	Ship Mode analysis	Profit margin by ship mode; shipping days vs. profitability	Understand cost-of-delivery implications
2G	Loss transaction deep-dive	Profile the ~18% of transactions with negative profit (what categories, discounts, regions?)	Root cause analysis for margin erosion

Expected critical finding — the discount cliff: Analysis of this dataset consistently reveals that transactions with discounts above ~20% have significantly reduced profit margins, and transactions with discounts above ~40% are almost universally unprofitable. This creates a clear, actionable recommendation about discount caps.

Pillar 3: Customer Churn Analysis & Prediction

Phase	Analysis	Method	Deliverable
3A	Customer lifetime profiling	Build customer-level feature table with all derived metrics	Model-ready dataset with 15+ features per customer
3B	Churn definition & rate	365-day inactivity threshold; calculate churn rate	Quantified churn rate with business justification
3C	Churn driver analysis	Grouped bar charts: churn by segment, discount level, categories purchased, order frequency	Visual identification of key churn correlates
3D	Model training	Logistic Regression, Random Forest, Gradient Boosting	3 trained models with AUC-ROC comparison
3E	Model evaluation	AUC-ROC, classification report, confusion matrix	Performance assessment

Phase	Analysis	Method	Deliverable
3F	Feature importance	Top 10 churn predictors from best model	Ranked, actionable list
3G	Segment-level churn	Churn rate comparison across Consumer, Corporate, Home Office	Segment-specific retention strategies

2.6 Dashboard Requirements

Page 1: Financial KPI Overview

Component	Type	Description
Total Sales ($)	KPI Card	Sum of all Sales
Total Profit ($)	KPI Card	Sum of all Profit
Overall Profit Margin	KPI Card	Total Profit / Total Sales × 100
Total Orders	KPI Card	Count of distinct Order IDs
Unique Customers	KPI Card	Count of distinct Customer IDs
Monthly Revenue & Profit Trend	Dual-Axis Line	Sales + Profit over time
Quarterly YoY Growth	Bar Chart	Year-over-year comparison by quarter
Filters	Interactive	Date range, Region, Category, Segment

Page 2: Profitability Deep Dive

Component	Type	Description
Sub-Category P&L	Horizontal Bar (diverging)	Sales and Profit side-by-side per sub-category, sorted by profit
Profit Margin Ranking	Lollipop / Bar Chart	Sub-categories ranked by profit margin %
Discount vs. Profit	Scatter Plot	Transaction-level Discount % (x) vs. Profit (y)
Discount Band Analysis	Grouped Bar	Avg profit margin by discount band (0%, 1–10%, 11–20%, 21–30%, 31%+)
State Profitability Map	Choropleth	U.S. map colored by profit margin per state

Component	Type	Description
Loss Leader Table	Table	Top 10 products by total negative profit

Page 3: Regional & Segment Performance

Component	Type	Description
Region Comparison	Bar Chart	Sales, Profit, Margin for East, West, Central, South
Top/Bottom States	Two-column table	5 most profitable + 5 least profitable states
Segment Comparison	Grouped Bar	Consumer vs. Corporate vs. Home Office on key metrics
Ship Mode Analysis	Bar + Margin Line	Sales and profit margin by shipping method
Category × Region Matrix	Heatmap	Profit margin by Category × Region

Page 4: Customer Retention & Churn

Component	Type	Description
Overall Churn Rate	Gauge	% of customers classified as churned
Revenue at Risk	KPI Card	Total Sales from churned customers
Churn by Segment	Bar Chart	Churn rate per customer segment
Churn by Purchase Frequency	Bar Chart	Churn rate bucketed by order count
Feature Importance	Horizontal Bar	Top 10 churn predictors from best model
Customer Value Distribution	Histogram	Total profit per customer — churned vs. active overlay

Page 5: Executive Recommendations

Component	Type	Description
Profit Opportunity Waterfall	Waterfall Chart	Current profit → potential improvement from each recommendation
Discount Policy Simulator	Interactive Sliders	"What if we cap discounts at 20%?" → projected profit impact

Component	Type	Description
Loss Sub-Category Decision	Cards	For each loss-making sub-category: options (fix pricing, reduce stock, exit)
Churn Recovery Opportunity	Calculator	Reactivation rate × avg customer value → recovered revenue
Action Items	Table	Specific recommendations, owners, timelines, estimated ROI

2.7 Expected Findings (Hypotheses to Validate)

Revenue & Growth

#	Hypothesis	Basis
H1	Revenue grew year-over-year from 2014 to 2017, with Q4 (Oct–Dec) being the strongest quarter each year	Holiday season retail demand + year-end corporate purchasing
H2	Growth is driven primarily by order volume rather than AOV increases	Typical pattern in mature retail businesses
H3	Technology generates the highest sales and profit per transaction, while Office Supplies drives the most volume	Technology has higher price points; Office Supplies are repeat consumables

Profitability

#	Hypothesis	Basis
H4	Tables, Bookcases, and Supplies are the three sub-categories operating at a net loss	Consistently reported in prior analyses of this dataset
H5	Transactions with discounts > 20% have significantly lower (or negative) profit margins	High discounts erode margin faster than they drive incremental volume
H6	Copiers is the most profitable sub-category by margin %, followed by Paper and Labels	High-value one-time purchases (Copiers) and low-cost consumables (Paper) have strong margins
H7	The Central and South regions have lower profit margins than East and West	Lower population density → potentially higher shipping costs and less competitive pricing
H8	Texas generates high revenue but relatively low profit (especially in Furniture)	Known pattern in this dataset — high sales volume with margin erosion

#	Hypothesis	Basis
H9	The Furniture category has the lowest overall profit margin of the three categories	High shipping costs for bulky items, heavy discounting on tables

Customer Churn

#	Hypothesis	Basis
H10	Customers with fewer than 3 total orders have significantly higher churn rates	Low-frequency buyers have weaker platform loyalty
H11	Customers who received higher average discounts are more likely to churn (trained to buy only on sale)	Discount-dependent customers are less loyal and more price-sensitive
H12	Customers who purchased across multiple categories are less likely to churn	Broader engagement = deeper relationship with the store
H13	Corporate and Home Office segments have lower churn than Consumer	B2B purchasing is more habitual and less discretionary
H14	Customer profit contribution (not just sales) is a stronger churn predictor than purchase frequency	Profitable customers receive better service/attention and have more satisfying experiences

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SECTION 3: EXPECTED OUTCOMES & BUSINESS IMPACT

3.1 Expected Financial Overview

Metric	Expected Value	Source
Total Sales (4 years)	~$2.30 million	Dataset aggregation
Total Profit (4 years)	~$286,000	Dataset aggregation
Overall Profit Margin	~12.5%	Calculated
Total Orders	~5,009	Distinct Order IDs
Unique Customers	793	Distinct Customer IDs

Metric	Expected Value	Source
Avg Order Value	~$459	Total Sales / Orders
Loss-Making Transactions	~18% of all transactions	Transactions where Profit < 0
Loss-Making Sub-Categories	3 (Tables, Bookcases, Supplies)	Sub-categories with total Profit < 0

3.2 Expected Model Performance

Revenue Forecast

Metric	Expected Value
Linear Trend R^2	0.70–0.85
YoY Revenue Growth Rate	15–20%
Strongest Seasonal Period	Q4 (Nov–Dec)

Churn Prediction Model

Model	Expected AUC-ROC
Logistic Regression	0.68–0.73
Random Forest	0.72–0.78
Gradient Boosting	0.75–0.80

3.3 Expected Sub-Category Profitability Ranking

Rank	Sub-Category	Expected Status	Expected Margin
1	Copiers	Highly Profitable	~35–40%
2	Paper	Profitable	~40–45%
3	Labels	Profitable	~35–40%
4	Phones	Profitable	~10–15%
5	Accessories	Profitable	~12–17%

Rank	Sub-Category	Expected Status	Expected Margin
6	Envelopes	Profitable	~25–30%
7	Art	Profitable	~15–20%
8	Binders	Marginally Profitable	~5–10% (heavy discounting)
...	...	...	...
15	Supplies	Loss-Making	~ –5 to –10%
16	Bookcases	Loss-Making	~ –5 to –15%
17	Tables	Loss-Making	~ –8 to –20%

3.4 Proposed Interventions & Expected Impact

Profitability Interventions

#	Intervention	Target	Expected Impact	Estimated Profit Improvement
1	**Cap maximum discount at 20%** — Eliminate discounts above 20% except for pre-approved clearance events	All transactions with Discount > 0.2 (~2,000 transactions)	Eliminate the deepest margin erosion; some volume may be lost but net profit improves	+$50,000–$80,000/year
2	**Exit or reprice Tables sub-category** — Either discontinue the lowest-margin table products or raise prices to achieve breakeven	Tables sub-category (net loss of ~$17,000 over 4 years)	Eliminate a persistent P&L drag	+$4,000–$6,000/year (loss elimination)
3	**Review Bookcases and Supplies pricing** — Investigate cost structure and vendor terms; raise prices or reduce product range	Bookcases and Supplies sub-categories	Move from loss-making to breakeven or marginally profitable	+$5,000–$10,000/year
4	**Regional profitability review for Texas, Ohio, Illinois** — Investigate why these states generate revenue but destroy profit; adjust discount policies, product availability, or shipping terms	State-level analysis	Targeted margin improvement in highest-loss states	+$10,000–$20,000/year

#	Intervention	Target	Expected Impact	Estimated Profit Improvement
5	**Promote high-margin sub-categories** — Increase marketing spend on Copiers, Paper, Labels, and Accessories which have the strongest margins	Merchandising and marketing teams	Shift product mix toward higher-margin items	+$15,000–$30,000/year

Churn Interventions

#	Intervention	Target	Expected Impact	Estimated Revenue Recovery
6	**Loyalty program for multi-category buyers** — Reward customers who purchase across 2+ categories	Customers with 2+ categories (~60% of base)	Deepen engagement, reduce churn risk	+$20,000–$40,000/year
7	**Reactivation campaign for lapsed customers** — Win-back offer for customers inactive 6–12 months	Customers approaching churn threshold	Re-engage before permanent churn	+$15,000–$25,000/year
8	**Corporate and Home Office retention program** — Dedicated account management and volume-based pricing	Corporate and Home Office segments	Reduce B2B churn, increase order frequency	+$30,000–$50,000/year

Financial Impact Summary

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DISCOUNT OPTIMIZATION:
  Current state: ~18% of transactions are unprofitable
  If discount cap at 20% eliminates 50% of loss-making transactions:
  ~900 transactions × avg loss of $60 × 50% recovery = $27,000/year

LOSS-MAKING SUB-CATEGORIES:
  Tables: ~-$17K over 4 years → ~-$4,250/year
  Bookcases: ~-$3K over 4 years → ~-$750/year
  Supplies: ~-$1K over 4 years → ~-$250/year
  Elimination of losses: +$5,250/year

CHURN RECOVERY:
  793 customers × estimated 20% churn rate = ~160 churned customers
  Reactivating 15%: 24 customers × $2,900 avg lifetime value = $69,600
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PRODUCT MIX SHIFT:
Shifting 5% of sales volume from low-margin to high-margin sub-categories:
$2.3M × 5% = $115K shifted × 20% margin improvement = $23,000/year

TOTAL ANNUAL PROFIT IMPROVEMENT OPPORTUNITY:
Conservative: $80,000–$120,000 (28–42% improvement on current $286K profit)
Aggressive: $150,000+ with full implementation

SECTION 4: DISCOUNT ANALYSIS FRAMEWORK

This section documents the discount-profit methodology — the most unique and impactful analytical component of this project.

4.1 Why Discount Analysis Is the Key Insight

Most retail analytics projects focus on revenue and customer behavior. This project's distinctive contribution is a rigorous analysis of the discount-to-profit relationship — essentially answering: **"Is our discounting strategy helping or hurting us?"**

This is a question that every CFO, category manager, and pricing analyst cares about. Demonstrating the ability to answer it with data is a powerful differentiator in interviews.

4.2 Analysis Framework

Step 1: Create Discount Bands

Band	Range	Expected % of Transactions
No Discount	0%	~48%
Light (1–10%)	1–10%	~5%
Moderate (11–20%)	11–20%	~30%
Heavy (21–30%)	21–30%	~10%
Extreme (31%+)	31–80%	~7%

Step 2: Analyze Each Band

For each discount band, calculate:

- Average profit per transaction
- Profit margin %
- % of transactions that are loss-making
- Total contribution to overall profit

Step 3: Identify the "Discount Cliff"

The point at which the average transaction transitions from profitable to unprofitable. Based on published analyses, this cliff typically occurs between **20–30% discount** for this dataset.

Step 4: Business Recommendation

Translate the finding into a specific, implementable policy: "Cap standard discounts at 20%. Any discount above 20% requires VP-level approval and must be tagged as clearance/liquidation."

SECTION 5: PROJECT DELIVERY & TIMELINE

5.1 Deliverables Checklist

#	Deliverable	Format	Status
1	This Project Report (BRD + PRD)	Markdown / PDF	✅ Complete
2	SQL Data Profiling & Analysis Queries	.sql files	⬜ To Build
3	SQL Revenue, Profitability & Churn Queries	.sql files	⬜ To Build
4	Data Exploration & Cleaning Notebook	Jupyter Notebook	⬜ To Build
5	Revenue Analysis & Forecasting Notebook	Jupyter Notebook	⬜ To Build
6	Profitability Analysis Notebook	Jupyter Notebook	⬜ To Build
7	Customer Churn Analysis & Prediction Notebook	Jupyter Notebook	⬜ To Build
8	Executive Summary & Recommendations Notebook	Jupyter Notebook	⬜ To Build
9	Tableau / Power BI Dashboard (5 pages)	.twbx / .pbix	⬜ To Build

#	Deliverable	Format	Status
10	GitHub Repository with README	GitHub	To Build
11	Executive Presentation (12-slide deck)	PowerPoint / PDF	To Build

5.2 Suggested Build Timeline

Week	Phase	Activities
Week 1	Data Acquisition & Profiling	Download dataset, load into SQL, profile all 21 columns, handle duplicates, validate data types and ranges
Week 2	EDA & Revenue Analysis	Monthly/quarterly trends, growth decomposition, seasonal patterns, category trends, build forecast model
Week 3	Profitability Analysis	Sub-category P&L, discount-profit analysis, regional profitability, segment comparison, loss transaction deep-dive
Week 4	Churn Analysis & Modeling	Customer-level feature engineering, churn definition, model training/evaluation, feature importance
Week 5	Dashboard & Presentation	Build 5-page Tableau dashboard, write business recommendations, financial impact estimates
Week 6	Polish & Package	GitHub repo, README, executive presentation, interview talking points, final QA

5.3 Tools & Technology Stack

Purpose	Tool
Data Profiling & SQL Analysis	SQLite / PostgreSQL
Data Analysis & Modeling	Python (Pandas, NumPy, Scikit-learn)
Visualization (Notebooks)	Matplotlib, Seaborn
Interactive Dashboard	Tableau Public or Power BI
Version Control	Git / GitHub
Documentation	Markdown / Jupyter Notebooks

5.4 Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Dataset is relatively small (9,994 rows) — may limit model complexity	Medium	Medium	Frame positively: "The dataset's clean structure and inclusion of profit data make it analytically richer than larger datasets without profit." Use simpler, well-validated models rather than deep learning.
Dataset is fictional/sample — interviewer may question its validity	High	Low	Prepare response: "This dataset is the industry-standard for Tableau and retail analytics demonstrations. I chose it because it includes actual profit data — enabling a level of financial analysis that most real publicly-available datasets cannot support. The methodology transfers directly to enterprise data."
Churn model may have limited predictive power with 793 customers	Medium	Medium	Focus the churn narrative on exploratory analysis (what drives churn) rather than model accuracy. Feature importance insights are valuable regardless of AUC.
Discount analysis may seem simple — "just cap discounts"	Medium	Low	Go deeper: show the discount cliff curve, quantify the specific dollar impact, and break the analysis down by sub-category (discounts may be appropriate for some products but destructive for others).
Multi-line orders require aggregation before customer-level analysis	High (first time)	High	Validate row counts at every aggregation step. One Order ID = 1–multiple line items. Customer ID is consistent and safe for groupby operations.

SECTION 6: APPENDIX

A. References

1. Superstore Sales Dataset — Kaggle. https://www.kaggle.com/datasets/vivek468/superstore-dataset-final
2. Tableau Sample Superstore Data — Tableau Public resources.
3. IEEE — "Predictive Customer Churn Analysis and Retention Strategy Development Using the Global Superstore Dataset" (2024).

4. Chen, D. et al. (2012). "Data mining for the online retail industry" — referenced for RFM methodology context.

B. Glossary

Term	Definition
Profit	Gross profit per transaction = Sales revenue minus cost of goods sold (COGS). Can be negative.
Profit Margin	Profit / Sales × 100. Expressed as a percentage.
Sales	Revenue from a transaction after discount has been applied.
Discount	Percentage reduction from list price (0.0 = full price, 0.2 = 20% off).
AOV	Average Order Value — total sales / number of orders.
Churn	When a customer stops purchasing. Defined here as no purchase in the last 365 days.
Sub-Category P&L	Profit & Loss statement at the sub-category level — aggregated sales minus aggregated costs.
Discount Cliff	The discount percentage threshold beyond which transactions become unprofitable on average.
Ship Mode	Method of shipping: Standard Class, Second Class, First Class, Same Day.
Segment	Customer type: Consumer (B2C), Corporate (B2B), Home Office (small business).
Region	Geographic division: East, West, Central, South.
AUC-ROC	Area Under the Receiver Operating Characteristic Curve — measures classification model quality.
Feature Engineering	Creating new analytical variables from raw data.
Loss Leader	A product sold at a loss to attract customers who then buy profitable items.

C. Cross-Project Portfolio Summary

Project	Domain	Dataset	Geography	Key Skill Demonstrated
1. Healthcare Analytics	Healthcare	UCI Diabetes 130 Hospitals (101K encounters, 50 features)	USA (130 hospitals)	Classification on imbalanced clinical data, feature engineering from ICD-9 codes

Project	Domain	Dataset	Geography	Key Skill Demonstrated
2. Retail & Marketing	Retail	UCI Online Retail (500K+ transactions, 8 columns)	UK + 38 countries	RFM segmentation, K-Means clustering, cohort retention analysis
3. Financial Operations	**Finance / Retail**	**Superstore (9,994 transactions, 21 columns incl. Profit)**	**USA (50 states)**	**Revenue forecasting, discount optimization, profitability P&L, churn prediction**

Portfolio progression narrative:

- Project 1 → Clinical domain, imbalanced classification, single-table analysis
- Project 2 → Customer segmentation, unsupervised ML, cohort analysis, single-table analysis
- Project 3 → Financial operations, profitability analysis with real P&L data, multi-pillar integrated analytics, U.S. market focus

All three projects follow the same **Business Problem → Data → Insight → Impact** framework, demonstrating consistent analytical maturity across domains.

D. Interview Talking Points

"Tell me about this project"

> "I built a Financial Operations Analytics Suite for a U.S. retail superstore, analyzing nearly 10,000 transactions across 4 years and all 50 states. What made this project distinctive is that the dataset includes actual profit per transaction — so I could do genuine P&L analysis, not just revenue tracking. I found that despite $2.3 million in sales, profit was only $286K — a 12.5% margin — and three sub-categories were operating at a net loss. My discount analysis revealed a 'discount cliff' at 20%: beyond that threshold, transactions became unprofitable on average. I also built a churn prediction model and found that customers who purchased across multiple categories were significantly less likely to churn. The key recommendation — capping standard discounts at 20% — alone could improve annual profit by $50K–$80K."

Follow-up: "Why this dataset over something larger?"

> "I chose it specifically because it includes actual profit data, which is rare in publicly available datasets. Most portfolio projects can only analyze revenue — they can tell you what sold the most but not what was actually profitable. The inclusion of profit, discount, region, and customer segment data let me build a genuine financial operations analysis. The methodology scales directly to enterprise data — the same discount-profit analysis, sub-category P&L, and churn modeling would work on a dataset with millions of rows."

Follow-up: "What was the most surprising finding?"

> "The discount cliff. Nearly half of transactions had no discount and were profitable. But once discounts exceeded 20%, profit margins collapsed. And at 40%+ discounts, virtually every transaction was a loss. That insight reframed the entire discount strategy from 'how much should we discount' to 'where should we draw a hard line.' The business was essentially giving away margin on transactions that probably would have happened at a smaller discount."